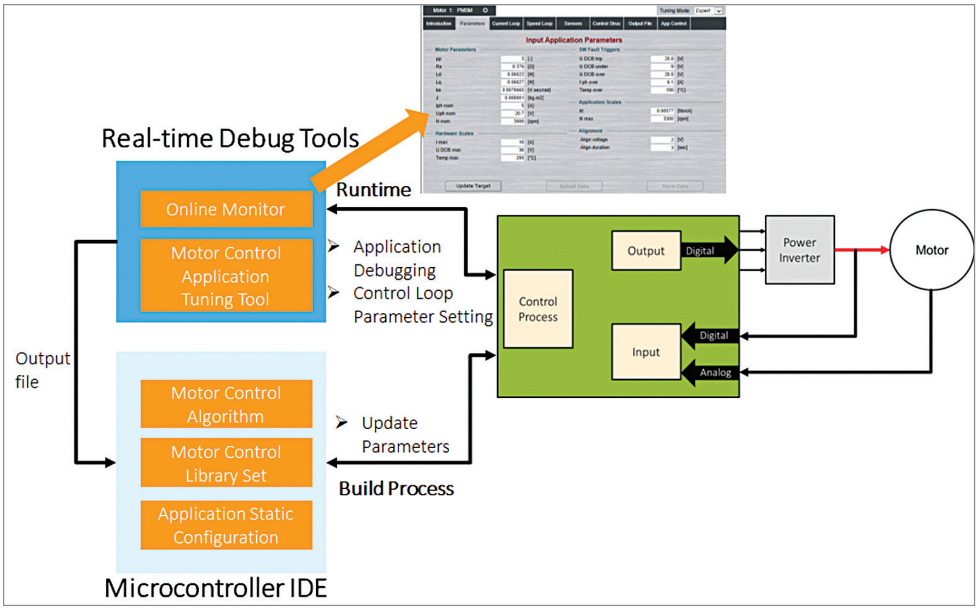

Fig. 4: Real-time debug tools are vital for quick tuning of motor control algorithm to specific motor

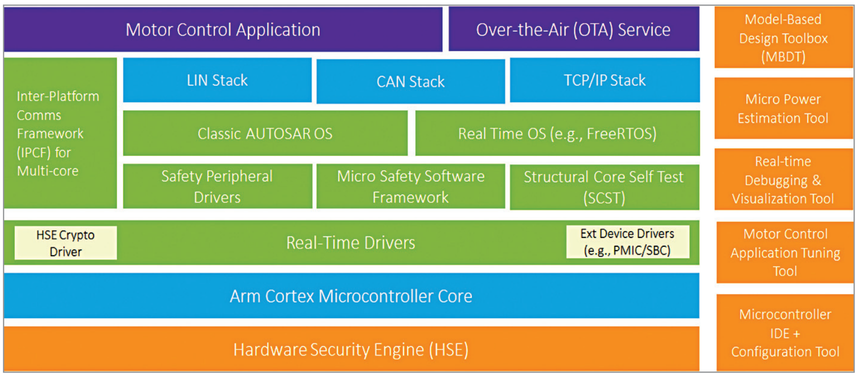

Fig. 5: A top-level view into a modern microcontroller software architecture in the context of electric 2W/3W motor control

process for motor control must execute as fast as once every 50µs.

One of the key challenges with developing a motor controller can be the complexity associated with the real-time control process and its associated application-level firmware development. This challenge can be broken down into a few specific problem statements under the overarching constraint of developing an optimised motor controller with very limited R&D resources and a fast time to market:

- Developing a motor control process algorithm
- Translating the motor control process algorithm into executable firmware on the microcontroller
- Calibrating and fine-tuning the control process algorithm to suit the target motor.

In this article, we will explore how model-based design (MBD), automotive math, and motor control libraries, along with real-time debug tools, address these challenges and enable a fast time-to-market for startups and established teams alike. The key enabler for this framework is also adopting an automotive microcontroller that supports a modular firmware architecture based on the modern ARM architecture.

Model-based design

Model-based design (MBD), which supports automatic code generation, enables rapid prototyping from idea incubation through to the final product. Robust support for MBD can be particularly useful for developing advanced features such as hill-hold and park-assist.

Automotive math and motor control libraries

A key step in translating the control process developed using MBD into executable firmware is the availability of automotive-quality math and motor control libraries. These perform best if available as bit-accurate models for the MBD toolbox and C-code for developers who prefer to develop and optimise their control process firmware in the microcontroller IDE. Ideally, the library is available in a hierarchical fashion, providing optimised functions not only for Park/Clark transformations but also for sine/cosine functions so that developers can have granular control over their control process and its optimisation.

Real-time debug tools for motor control

Motor controller developers will inevitably find the need to fine-tune their control process to suit specific motor types. It can take days to sometimes even weeks of bench-top effort to iteratively estimate optimised control loop parameters that are perfectly matched for the motor. Debugging can be cumbersome in this stage of development. Here, developers would do well to pay careful attention to the easy availability of real-time debug tools that provide the following key functions:

- Run-time configuration and tuning of control processes in the application firmware